



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

– Hira Kumar

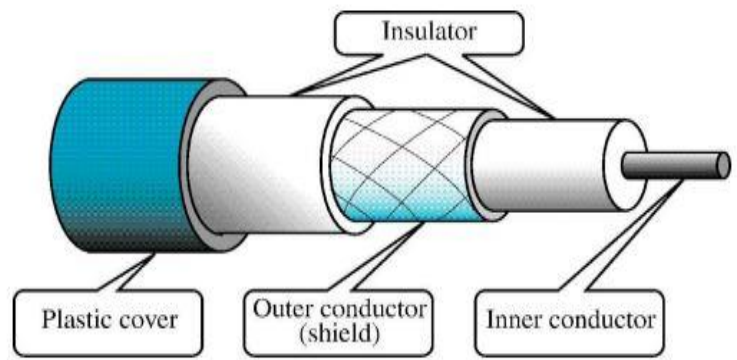
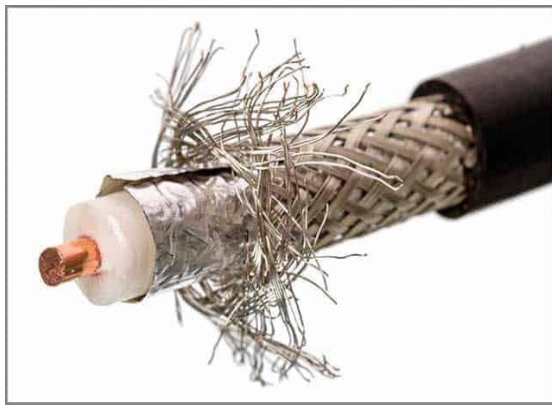
Computer Networking



Data Communication

B. Co-axial cable

- The name "coaxial" comes from the fact that it contains two conductors that share a common axis.
- The inner conductor is made of copper and is surrounded by PVC / Teflon insulation.
- The outer conductor is made of metal foil(पत्ति),, a metal mesh (braid) (जाली), or both.
- The outer metallic conductor is used as a shield against noise and electromagnetic interference (EMI).
- The outermost part is a plastic cover that protects the entire cable.
- Coaxial cable is much less susceptible(अतिसंवेदनशील) to interference and crosstalk than twisted-pair cable.
- Coaxial cable is used to transmit both analog and digital signals.



➤ Application /Use

- ✓ TV distribution cable /Cable TV – Present time
- ✓ Long distance telephone transmission – Not at the present time
- ✓ Short distance communication links (pc to pc) – Not at the present time
- ✓ LAN - Not at the present time
- ✓ CCTV – Present time
- ✓ Cable modem internet - present time

➤ Co-axial cables usually for data transmission are of two types,its called Ethernet network standards.

Standard	Cable used	Nickname	Max length
10BASE5	Thick Coaxial cable (RG-8)	Thicknet	500m
10BASE2	Thin coaxial cable (RG-58)	Thinnet	200m

here

10 = 10 Mbps speed

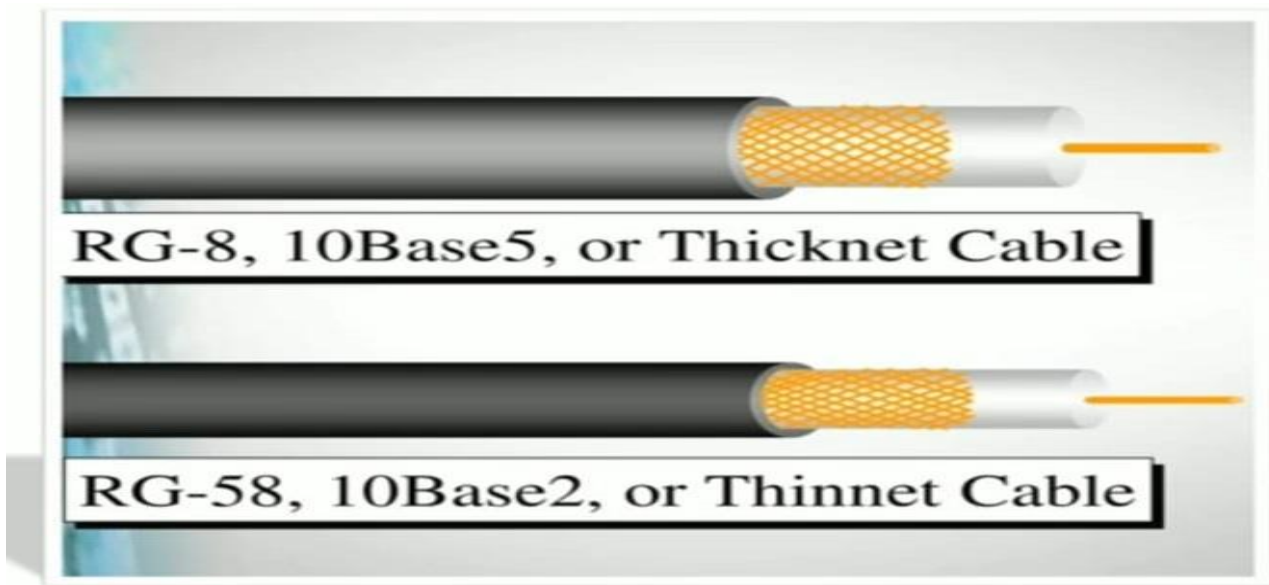
BASE = Baseband transmission (Digital signal is transmitted directly without modulation)

5 = Approx. 500m for reliable data transmission

2 = Approx. 200m for reliable data transmission

What is Modulation : The process of combining a message signal with a carrier signal for transmission.

What is carrier signal : A high frequency signal used to carry information from one place to another.



➤ **BaseBand**

- It is mostly used for LANs (local area network).
- Baseband transmits only one signal at a time at high speed.
- In traditional baseband coaxial cable, a repeater is required approximately every 1000 feet (300 meters) to regenerate the digital signal.
- No modulation
- Use : Ethernet LAN

➤ **BroadBand**

- It uses analog transmission over standard cable television (CATV) cables.
- It transmits multiple signals simultaneously using different frequencies.
- It covers a larger area than baseband coaxial cable.
- Modulation required
- Use : TV, Radio, DSL(digital subscriber line), Mobile network

Quick Difference

Baseband

Digital transmission

One signal at a time

Mostly used in LAN

Uses repeaters for long distances

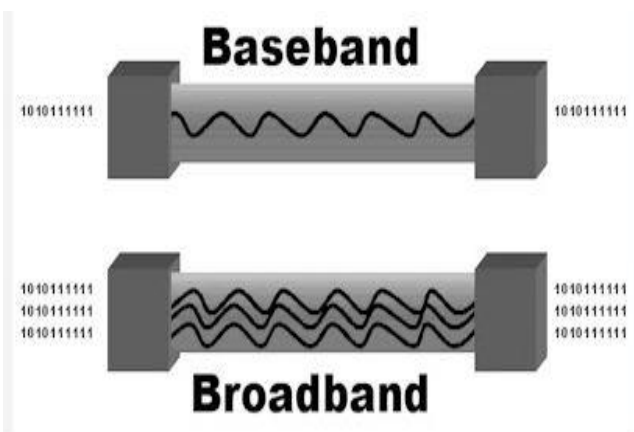
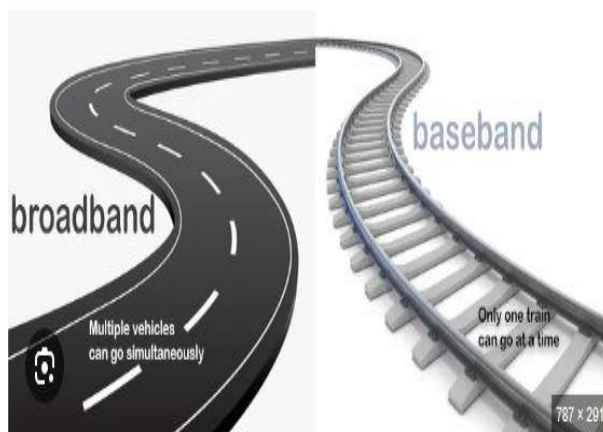
Broadband

Analog transmission

Multiple signals at the same time

Used in Cable TV and Cable Internet

Uses amplifiers for analog signals



➤ Bandwidth

- The Bandwidth is the maximum amount of data that can be transmitted over a network connection in a given amount of time.
- In other words, bandwidth is the maximum data transfer capacity of a communication channel.
- Bandwidth is usually measured in Mbps (Megabits per second) or Gbps (Gigabits per second).



➤ Throughput

- Throughput is the actual amount of data successfully transferred from one end to another in a given amount of time.
- Example: If the bandwidth is 30 Mbps, but only 25 Mbps is actually received, then the throughput is 25 Mbps.

➤ Latency

- Latency is the time taken for data to travel from the source to the receiver.
- It depends on factors such as network traffic, transmission medium, and distance.
- Latency is measured in milliseconds (ms).

All Example :

Bandwidth: 100 Mbps (maximum capacity) [data 100mpbs के स्पीड से भेज सकता हैं }

Throughput: 80 Mbps (actual speed achieved) {data 80 mbps पहुँचा }

Latency: 20 ms (time taken for data to reach the destination) {समय 20मिलीसेकंड लिया}

Extra Topic (Not in Syllabus)

Twisted Pair cable

Cable name	Connector name	use
RG-6	F-type connector	Cable tv, cable broadband internet
RG-59	BNC connector	Analog CCTV camera
RG-58	BNC connector	Old ethernet LAN (10BASE2), radio
RG-11	F-type connector	Long-distance cable tv/ broadband



F-TYPE

BNC



VS

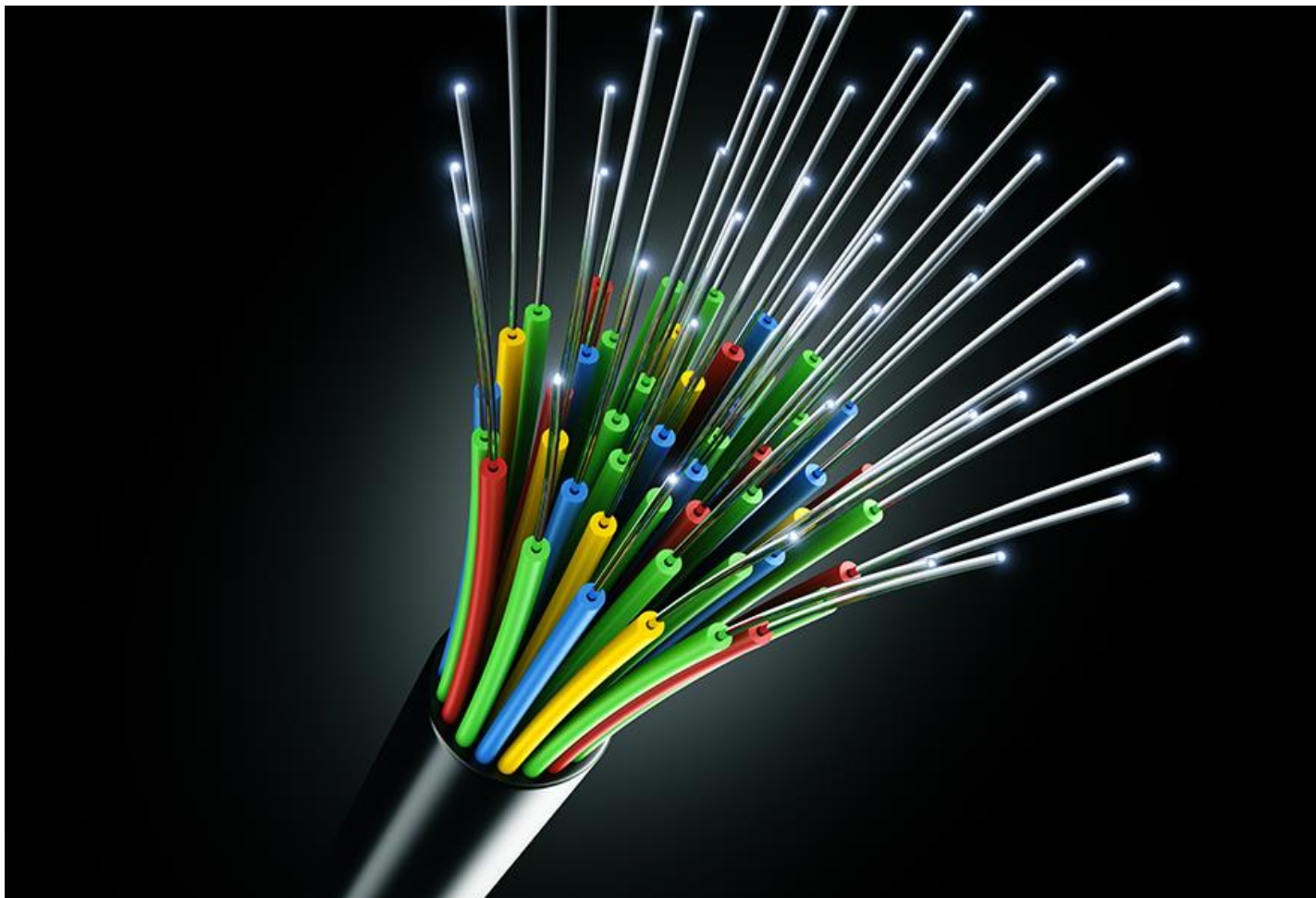


75 OHMS
CABLE TV, SATELLITE

50 OHMS
VIDEO, RF

C. Fiber-optic cable OR Optical Fiber

- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
- It works on the principle of **Total internal reflection (TIR)**
- A light pulse represents a binary 1, while the absence of a light pulse represents a binary 0.
- The bandwidth of an optical transmission system is very high.
- It supports extremely high-speed data transmission.
- The transmission medium is an ultra-thin fiber made of glass or plastic.

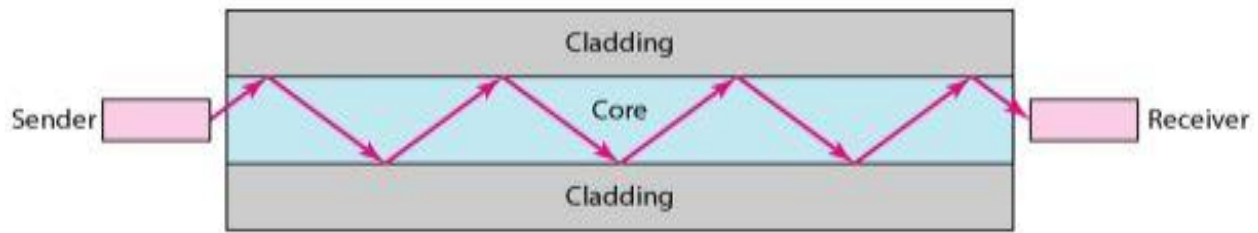


An optical fiber has a cylindrical shape and consists of three concentric sections:

(i) Core

(ii) Cladding

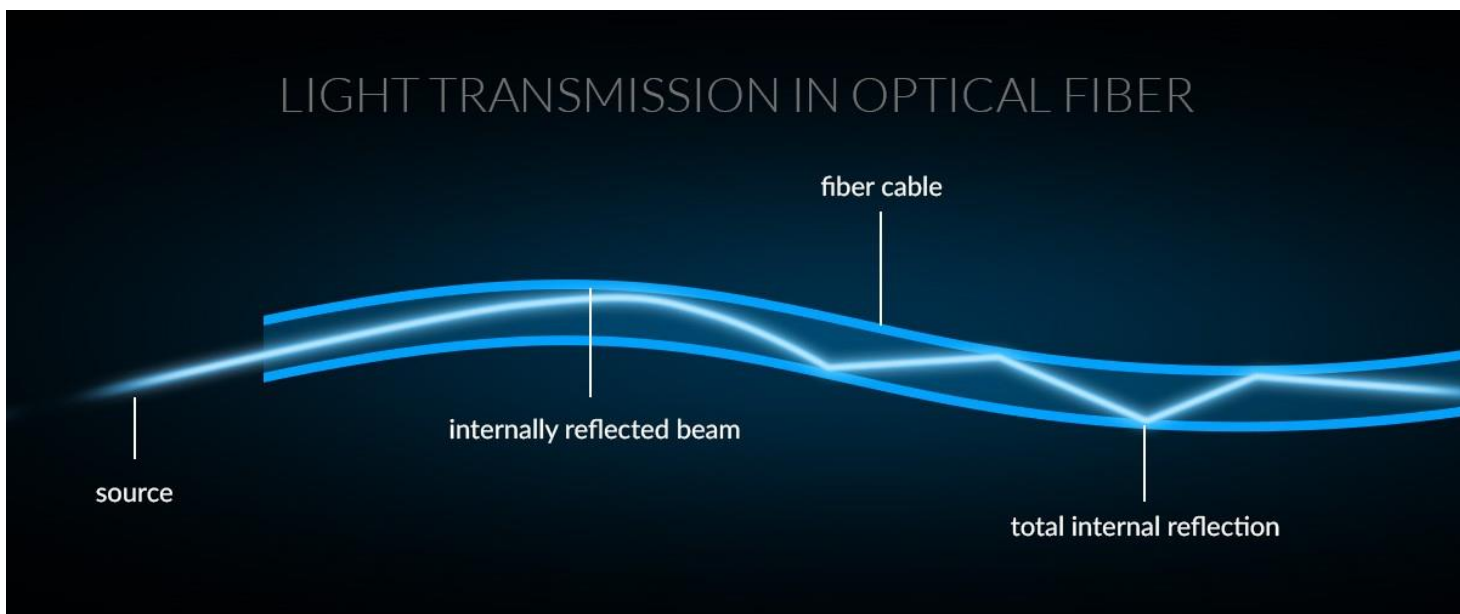
(iii) Jacket



- 1. Core:-** It's the inner most section is made of glass or plastic and is surrounded by its own cladding. The core diameter is in the range of 8 to 50 μm .
- 2. Cladding:** - A glass or plastic coating that has optical properties different from those of the core having a diameter of 125 μm . The cladding acts as a reflector to light that would otherwise escape the core.
- 3. Jacket:** - The outer most layers surrounding caddied fiber is the jacket. Jacket is composed of plastic or other material layer to protect against moisture, cut, crushing and other environmental dangers.

➤ **Optical Fiber Communication**

- A transmitter (light source) at the sender's end sends light through the optical fiber.
- A receiver at the other end uses a light-sensitive device to detect the presence or absence of light, indicating binary 1 or 0.
- **Total Internal Reflection (TIR)** is the complete reflection of light back into a denser medium when it travels from a denser medium to a rarer medium at an angle greater than the critical angle.



➤ **Two types of light sources are commonly used in fiber-optic communication systems:**

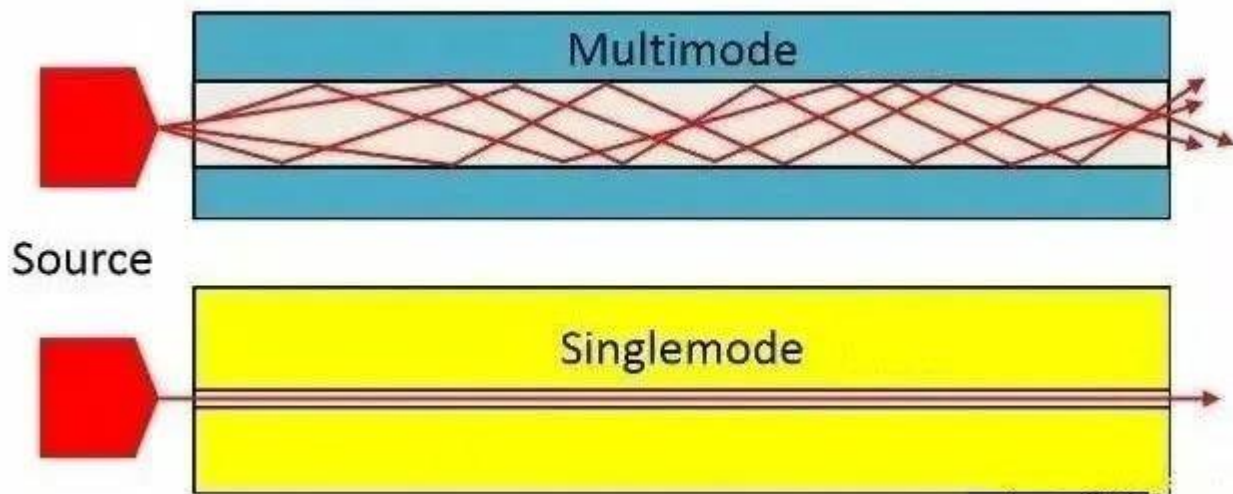
- The Light Emitting Diode (LED)
- Injection Laser Diode (ILD).

➤ **Application/Use**

- ✓ Telephones
- ✓ including cellular wireless Internet
- ✓ LANs - local area networks, WAN
- ✓ CCTV Camera
- ✓ Medical Application
- ✓ Transportation – smart lights and highways
- ✓ Military

✓ **Extra Topic (Not in Syllabus)**

Twisted Pair cable		
Cable name	Connector name	use
SMF (single-mode fiber)	SC,LC,FC	Long distance (telephone/ mobile network, ISP, WAN,home)
MMF (Multimode fiber)	SC,LC,ST	Short-distance, LAN, office, building, data center



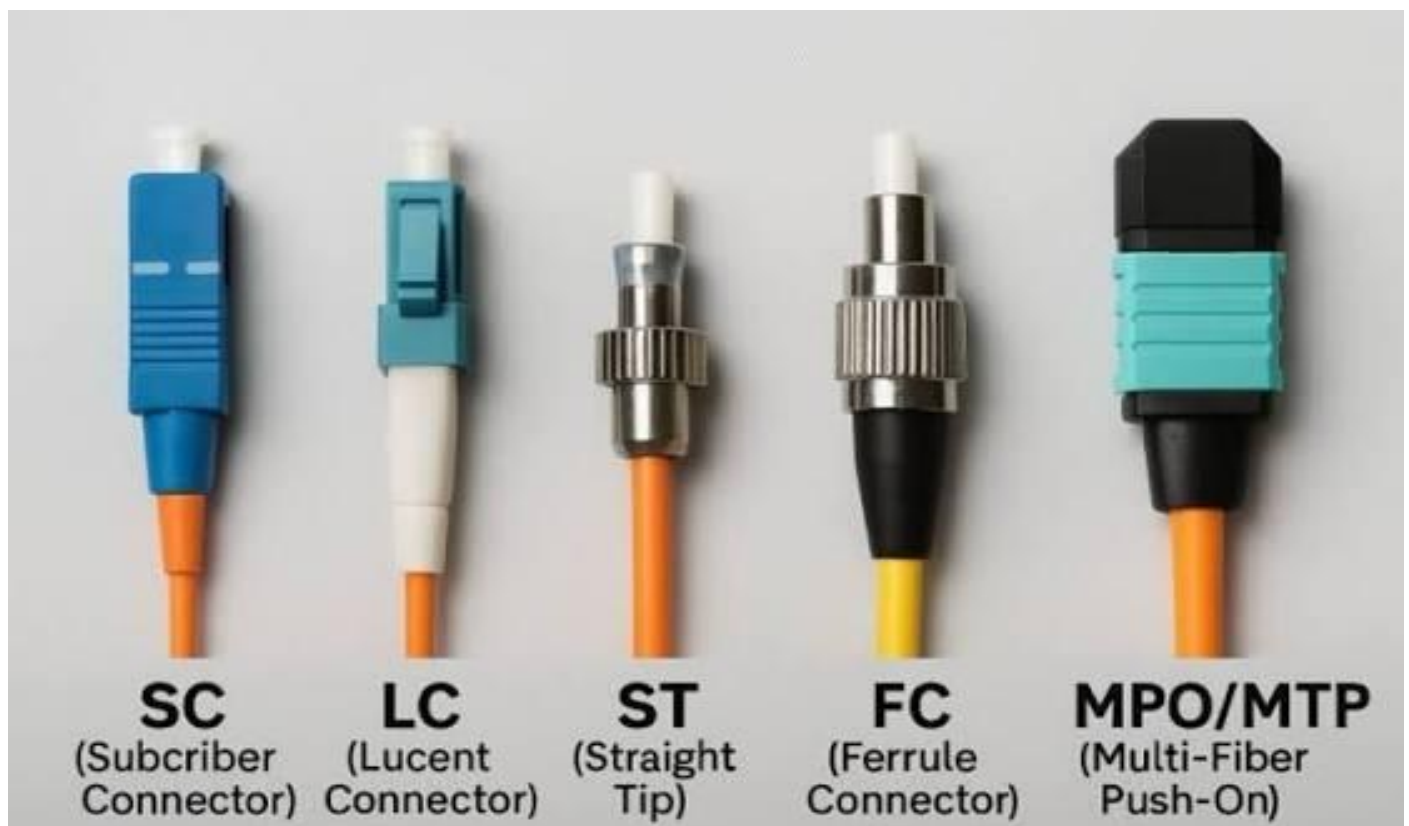
SMF



MMF



MMF



Practical

Video 1 = Fiber stripping (फाइबर के बाहरी पार्ट को हटाना)

<https://www.youtube.com/shorts/DrlaMo7MuLk>

Video 2 = 2 fiber cable join process

<https://www.youtube.com/shorts/-Fm7qwMklh0> (outside machine)

<https://www.youtube.com/shorts/YWqR3Ae3AZ4> (inside machine)

Video 3 = Inside optical fiber

<https://www.youtube.com/shorts/-xf1BY1L6r8>

Video 4 = Check fiber cable fault / damage

<https://www.youtube.com/shorts/D7BAKs4Xq4w>

<https://www.youtube.com/shorts/KCunTO9-5iA>

Video 5 = TIR video of fiber cable

<https://www.youtube.com/shorts/PhJ8uxlOqO4>

➤ **Difference / Characteristics of cable media:-**

DIFFERENCE BETWEEN UTP, STP, COAXIAL AND FIBER OPTIC CABLE

Feature	UTP CABLE (Unshielded Twisted Pair)	STP CABLE (Shielded Twisted Pair)	COAXIAL CABLE	FIBER OPTIC CABLE
 Cable Image				
 Construction	4 Pairs of copper wires twisted together	Twisted pairs of copper wires with overall shield (foil/braid)	Central copper conductor, dielectric insulation, metal shield, outer jacket	Glass or plastic core, cladding, protective coating, outer jacket
 Shielding	No Shielding	Shielded (Foil or Braid)	Shielded (Metal Braid/Shield)	No Shielding (Uses light)
 Medium / Signal	Electrical (Copper)	Electrical (Copper)	Electrical (Copper)	Light (Optical Fiber)
 Data Transmission	Digital	Digital	Analog / Digital	Digital (Light Pulses)
 Speed	Up to 1 Gbps (10 Gbps for Cat 6a/7)	Up to 1 Gbps (10 Gbps for Cat 6a/7)	Up to 100 Mbps – 1 Gbps (Depends on type)	Very High Speed (Mbps to Tbps)
 Distance	Up to 100 meters	Up to 100 meters	Up to 500 meters (Depends on type)	Up to 2 km to 100+ km (Depends on type)
 Interference (Immunity)	Low (More Affected by EMI)	High (Better protection from EMI)	High (Good protection from EMI)	Very High (Immune to EMI and RFI)
 Common Uses	LAN, Telephone, Networking	LAN, Industrial Networks, High EMI areas	Cable TV, Internet (Broadband), CCTV	Long Distance Communication, Internet Backbone, Data Centers
 Connector	RJ45	RJ45	BNC, F-Type, N-Type	SC, LC, ST, FC
 Cost	Low	Medium	Medium	High
SUMMARY	UTP is low cost, easy to install and used in LANs, but more susceptible to interference.	STP has shielding, better performance in noisy environments, used in LANs and industries.	Coaxial cable is durable, has good shielding and used in cable TV and broadband connections.	Fiber optic cable offers very high speed and long distance with no interference, used in advanced communication systems.

=====HIRA KUMAR=====